

Apple Intelligence in Flutter Apps

Bridging Dart and Apple's on-device Foundation Models through a native platform channel



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Mobile App Developer



What is Apple Intelligence?

What Is Apple Intelligence?

Apple's personal intelligence system, woven into iOS, iPadOS, and macOS



Writing Tools

Rewrite, proofread, and summarize text system-wide



Image Playground

Generate original images right on-device



Smarter Siri

Richer context and on-screen awareness



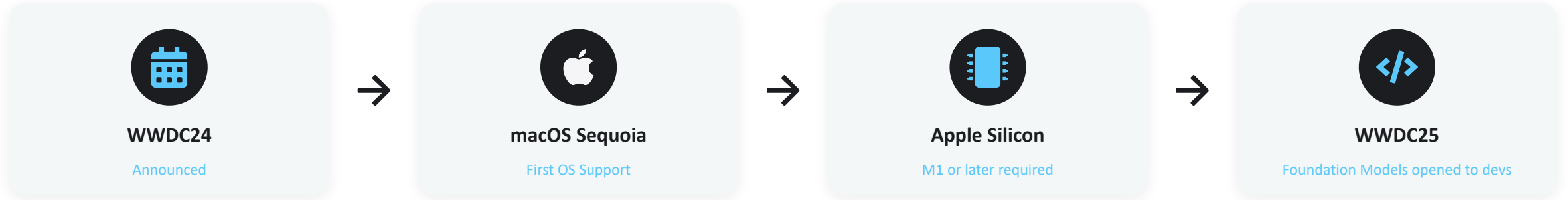
Priority & Summaries

Notifications and messages, distilled

All powered by on-device models first — Private Cloud Compute steps in only for harder tasks

How Apple Intelligence Reached the Mac

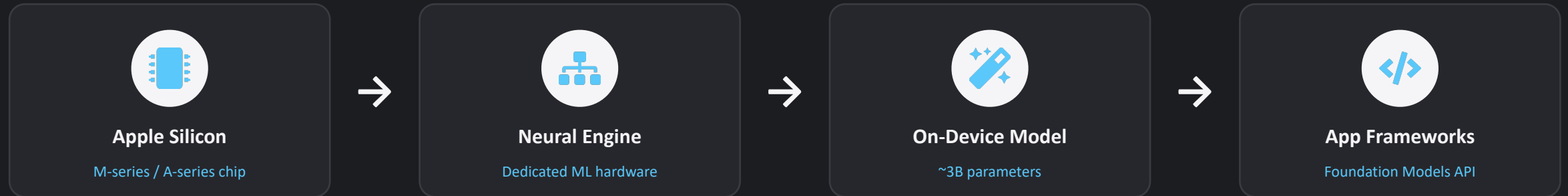
From an on-stage announcement to a public developer API



Today: any app — including Flutter apps via a native bridge — can call the same on-device model

How It Runs on Your Device

Every request starts on-device before ever considering the cloud



If a task needs more power, Private Cloud Compute runs it on Apple silicon servers — same privacy guarantees, verifiable, and never stored.



From Concept to Code

Now let's wire it into a real Flutter app

What You Get: Foundation Models

Apple's on-device LLM, exposed natively to Swift — no API keys, no server costs



Runs On-Device

Data never leaves the phone



Zero Cost

No API bills or rate limits



3B Parameters

Summarizing, extracting, classifying



Native Swift API

Reached from Flutter via a bridge

Size	Location	Context	Cost	Min. OS
~3B params	On-device	~4K tokens	Free	iOS 26+



Good at

- Summarizing, extracting, classifying text
- Structured output via Guided Generation



Not for

- General world knowledge or open chat
- Long documents or multi-turn deep reasoning

Requirements to Get Started

Everything that needs to be true before Apple Intelligence can be called — including from Flutter



Xcode Version

Xcode 26 or later — not available in Xcode 16.3



macOS Version

Sequoia 15.6+ to install the Xcode 26 base release



SDK Target

iOS 26 SDK as the minimum deployment target

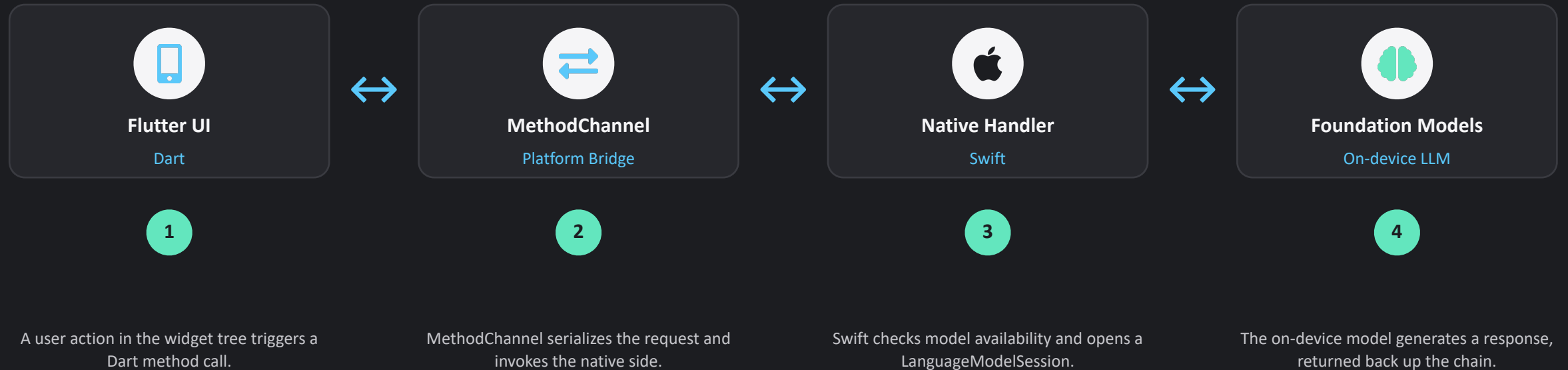


Test Hardware

A17 Pro / M-series chip with Apple Intelligence enabled

How Flutter Connects to Apple Intelligence

Dart can't call Foundation Models directly — a native platform channel bridges the two



Dart never talks to Foundation Models directly — the native Swift layer does all on-device inference.

Adding It to Our Project

Where Foundation Models plugs into an existing SwiftUI + SwiftData app

```
import FoundationModels

func generateInsight(from log: [HabitEntry]) async throws -> String {
    let model = SystemLanguageModel.default
    guard case .available = model.availability else {
        return "Insights unavailable on this device"
    }

    let session = LanguageModelSession()
    let prompt = "Summarize this habit streak: \(log)"
    let response = try await session.respond(to: prompt)
    return response.content
}
```

Call it from your ViewModel — same layer that already talks to SwiftData

The Dart Side of the Bridge

Calling into Swift with a typed MethodChannel

```
final _channel = const MethodChannel('com.app/foundation_models');

Future<String> generateInsight(List<HabitEntry> log) async {
  try {
    final result = await _channel.invokeMethod<String>(
      'generateInsight',
      {'log': log.map((e) => e.toJson()).toList()},
    );
    return result ?? 'No insight available';
  } on PlatformException catch (e) {
    return 'Insight unavailable: ${e.message}';
  }
}
```

Pigeon can generate this boilerplate + type-safe models automatically



Live Demo

Running this on macOS and the iOS Simulator



Trigger from the Flutter UI



MethodChannel bridges to
Swift



Foundation Models responds
on-device

Why On-Device Instead of a Cloud LLM

Foundation Models vs. a cloud LLM API vs. task-specific on-device ML

Aspect	Foundation Models	Cloud LLM API	On-device ML Kit
Cost	Free	Pay-per-token	Free
Privacy	Stays on device	Sent to a server	Stays on device
Latency	Fast, local	Network dependent	Fast, local
Offline	Works	Fails	Works
Capability	Good for short tasks	Broad, powerful	Narrow, task-specific
Footprint	OS downloads the model — device needs the space	No download, zero footprint	Bundled in app — app size gets heavy



What Can We Build?

A few ideas worth stealing for your next feature

Use Cases We Can Build



On-Device Chat

Conversational Q&A that runs fully offline, no server round-trip



RAG over Local Data

Ground answers in on-device notes, files, or history



Smart Summarization

Condense long text into key points in seconds



Content Classification

Auto-tag or categorize text by topic or intent



Text Rewriting

Polish tone, grammar, and clarity system-wide



Structured Extraction

Pull typed data out of free text via @Generable



Recap

- 1 Foundation Models is free, private, and fully on-device — but Dart can't call it directly
- 2 A native platform channel (or Pigeon) bridges Flutter to Swift's Foundation Models API
- 3 Best fits: summarizing, tagging, and short nudges — not general chat or deep reasoning



Resources

- [Apple Developer Documentation — Foundation Models framework](#)
- [Flutter Documentation — Platform Channels](#)
- [WWDC session on Foundation Models and Guided Generation](#)



Questions?

Happy to go deeper on the bridge, the model, or the architecture

Thank You

Let's Connect



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